

Def. Let G be a group, and $H \leq G$ be a subgroup.
 Given an element $g \in G$, define left-Coset of H as

$$gH \stackrel{\text{def}}{=} \{gh \mid h \in H\}$$

Lemma. Let G be a group, and $H \leq G$ be a subgroup. Then, for each $g_1, g_2 \in G$,
 $g_1H = g_2H$ if and only if $g_1^{-1}g_2 \in H$.

Proof. Given $g_1, g_2 \in G$, suppose that $g_1H = g_2H$. Then, since $g_2 \in g_2H = g_1H$ ($1 \in H$),
 $g_2 = g_1h$ for some $h \in H$, thus $g_1^{-1}g_2 \in H$.

Conversely, let $g_1, g_2 \in G$ be given s.t. $g_1^{-1}g_2 \in H$. That is, $g_1h = g_2$ for some $h \in H$.
 Given $g_1h_1 \in g_1H$, $g_1h_1 = g_2h^{-1}h_1 \in g_2H$. Given $g_2h_2 \in g_2H$, $g_2h_2 = g_1hh_2 \in g_1H$ $\square$

Q. In left Coset sense, $g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H$ hold?

A. No!, s.t. $H = \langle (1, 2) \rangle$, $g_1 = (1, 3)$ and $g_2 = (1, 2, 3)$

Thm. Lagrange Thm

If H is a subgroup of a group G , then $|G|/|H| = [G:H]$

Proof. Given a subgroup $H \leq G$, define a relation on G as;

$$x \sim y \Leftrightarrow xH = yH$$

Then, it is an equivalence relation, so a Collection $\{gH \mid g \in G\}$ forms a partition.
 Moreover, for each $g \in G$, a map

$$H \rightarrow gH : h \mapsto gh$$

is onto by definition and one to one by the cancellation law, so $|H| = |gH|$.